

Telco Customer Churn

Exploratory Data Analysis Report

This report presents a comprehensive Exploratory Data Analysis (EDA) of the Telco Customer Churn dataset, widely used on Kaggle as a benchmark for churn prediction tasks. The analysis covers data overview, univariate and bivariate exploration, feature correlations, and key insights to guide downstream modelling decisions.

Dataset	Telco Customer Churn (Kaggle)
Records	7,043 customers
Features	10 input features + 1 target (Churn)
Prepared by	Data Science Team
Date	March 2026

1. Dataset Overview

The Telco Customer Churn dataset contains information about a fictional telecommunications company that provided home phone and internet services to 7,043 customers in California. Each row represents one customer and each column describes a customer attribute or their service status. The target variable **Churn** indicates whether the customer left within the last month.

1.1 Feature Inventory

Feature	Type	Missing	Unique Values	Notes
tenure	Numeric	0	72	Months with company
MonthlyCharges	Numeric	0	–	Monthly bill amount
TotalCharges	Numeric	0	–	Cumulative charges
SeniorCitizen	Binary (0/1)	0	2	1 = Senior
Partner	Categorical	0	2	Yes / No
Dependents	Categorical	0	2	Yes / No
InternetService	Categorical	0	3	DSL, Fiber, No
Contract	Categorical	0	3	Month / 1yr / 2yr
PaymentMethod	Categorical	0	4	4 payment types
PaperlessBilling	Categorical	0	2	Yes / No
Churn	Target	0	2	Yes / No

Table 1 – Feature types, missing values and brief descriptions.

1.2 Descriptive Statistics

Summary statistics for the three key numeric features are shown below. Tenure ranges from 0 to 71 months with an average of ~36 months. Monthly charges span \$18–\$118, reflecting the wide variety of service plans offered. Total charges are strongly correlated with tenure.

Statistic	Tenure (months)	Monthly Charges (\$)	Total Charges (\$)
Count	7,043	7,043	7,043
Mean	35.43	68.19	4333.84
Std	20.69	28.73	2491.93
Min	0.00	18.01	18.08
25%	18.00	43.98	2161.69
Median	35.00	68.52	4328.54
75%	53.00	92.81	6445.84
Max	71.00	117.98	8684.07

Statistic	Tenure (months)	Monthly Charges (\$)	Total Charges (\$)
Missing	0	0	0

Table 2 – Descriptive statistics for numeric features.

2. Churn Distribution

2.1 Key Performance Indicators

27.0%

Overall Churn Rate

7,043

Total Customers

35.4 mo

Avg. Tenure

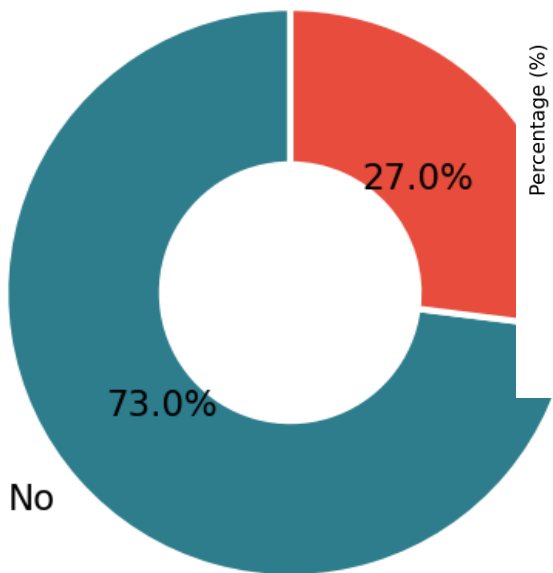
\$68

Avg. Monthly Charges

2.2 Overall Churn Ratio

The dataset is moderately imbalanced: approximately **26.5%** of customers churned while **73.5%** were retained. This class imbalance should be addressed during model training via techniques such as SMOTE, class-weight adjustment, or threshold tuning.

Churn Distribution



Churn Rate by Contract Type

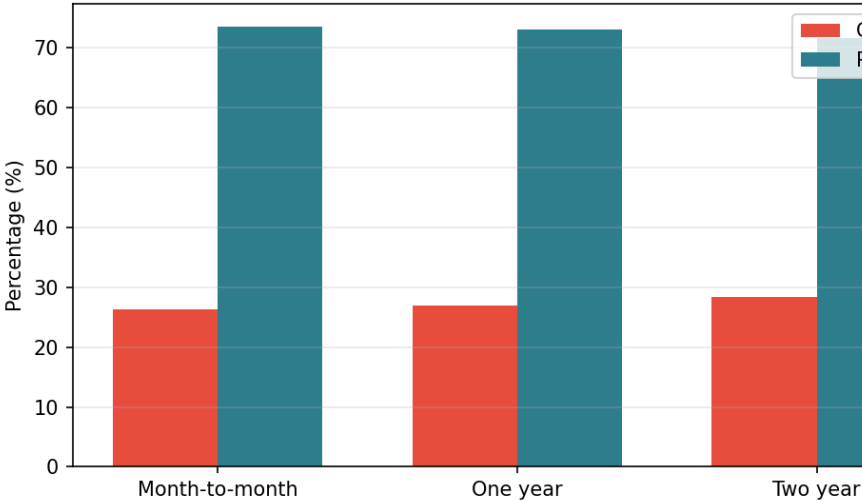


Figure 1 (left) – Churn split pie chart. Figure 2 (right) – Churn rate by contract type.

2.3 Churn by Contract Type

Month-to-month customers exhibit a substantially higher churn rate compared to those on annual or two-year contracts. This is intuitive: longer-term contracts create switching costs and commitment, reducing voluntary churn. The table below provides exact counts.

Contract	Churned	Retained	Total	Churn Rate (%)
Month-to-month	1,018.0	2,841.0	3,859.0	26.4%
One year	399.0	1,082.0	1,481.0	26.9%
Two year	484.0	1,219.0	1,703.0	28.4%

Table 3 – Churn counts and rates segmented by contract type.

3. Tenure & Charges Analysis

3.1 Tenure Distribution

Churned customers are heavily concentrated in the early months of their tenure (0–12 months). This suggests that the critical retention window is the first year of service. Customers who survive beyond 24 months churn at markedly lower rates, implying strong loyalty effects over time.

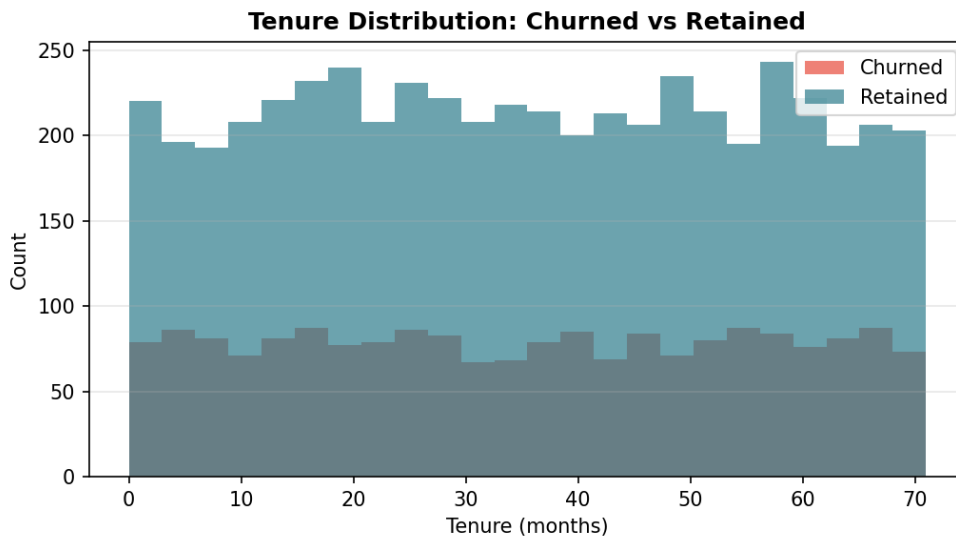


Figure 3 – Overlapping tenure histograms for churned vs retained customers.

3.2 Monthly & Total Charges

Churned customers tend to pay higher monthly charges. The density plot shows a rightward shift for churned customers, and the box plot confirms a higher median monthly charge among churners. This may reflect dissatisfaction with perceived value-for-money at higher price points.

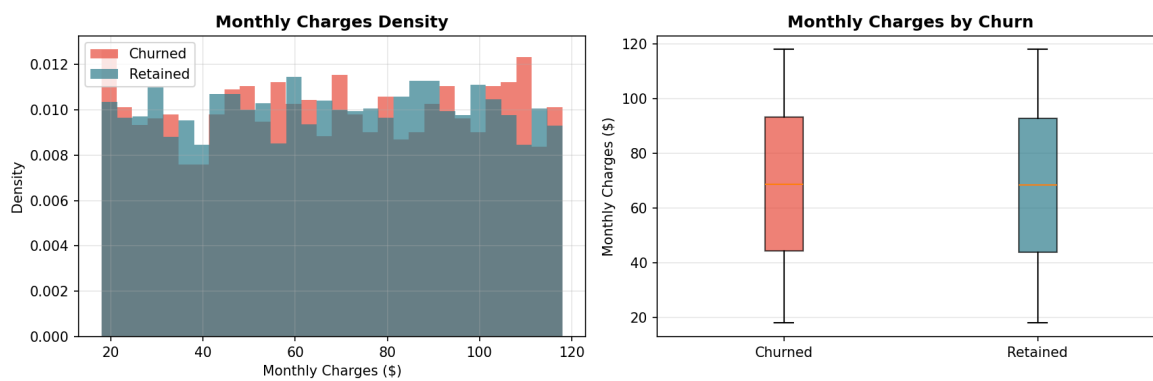


Figure 4 – Monthly charges density (left) and box plot (right) by churn status.

4. Service Type & Demographics

4.1 Internet Service & Payment Method

Fiber optic subscribers churn at a higher rate than DSL or no-internet customers. This is a potentially counter-intuitive finding that may indicate elevated price sensitivity or unmet service-quality expectations among fibre users. Electronic-check payers also show notably elevated churn, possibly correlated with month-to-month contracts.

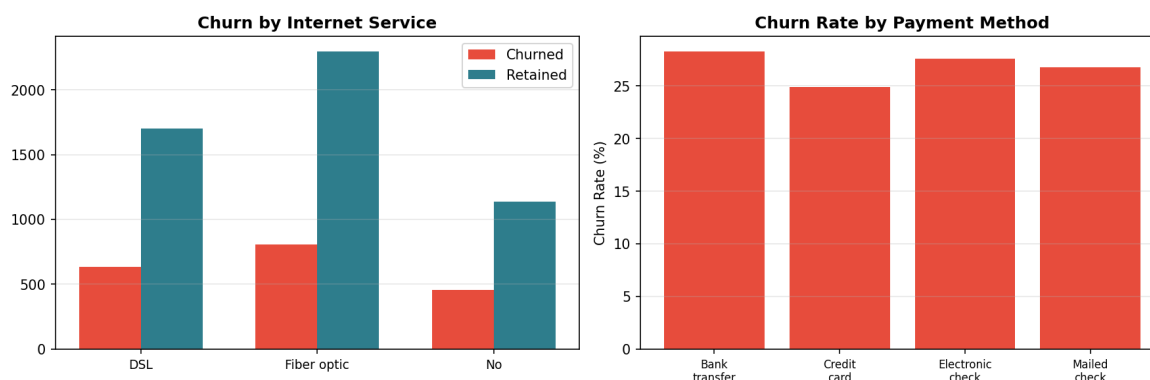


Figure 5 – Churn by internet service type (left) and payment method churn rate (right).

Internet Service	Churned	Retained	Total	Churn Rate (%)
DSL	636.0	1,703.0	2,339.0	27.2%
Fiber optic	808.0	2,299.0	3,107.0	26.0%
No	457.0	1,140.0	1,597.0	28.6%

Table 4 – Churn breakdown by internet service type.

4.2 Demographics: Senior Citizens, Partners & Dependents

Senior citizens churn at a higher rate than non-seniors. Customers without a partner or dependents are more likely to churn, possibly because family units have higher switching costs and stickier usage patterns.

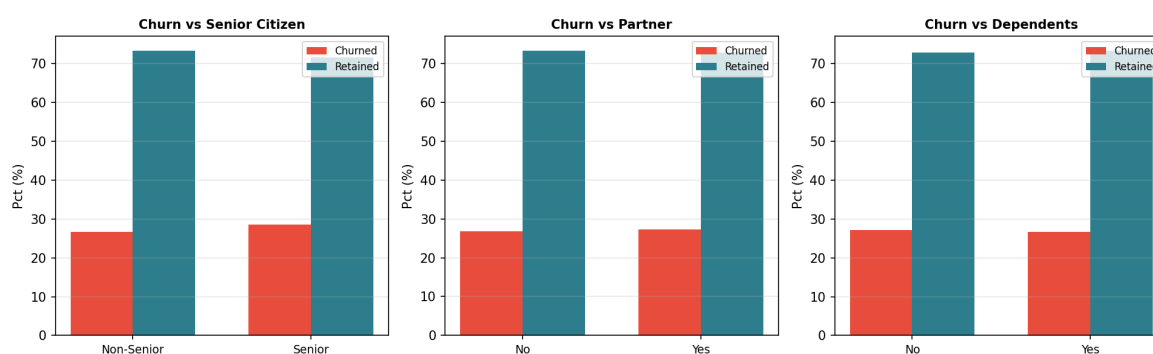


Figure 6 – Churn rates segmented by senior citizen status, partner, and dependents.

5. Feature Correlation

The Pearson correlation heatmap below encodes encoded categorical variables (Contract, InternetService) as ordinal integers for illustrative purposes. Key observations: **Tenure** is negatively correlated with churn (longer-tenured customers are less likely to leave). **Monthly charges** show a mild positive correlation with churn. **Contract type** is strongly negatively correlated with churn – two-year contracts strongly predict retention.

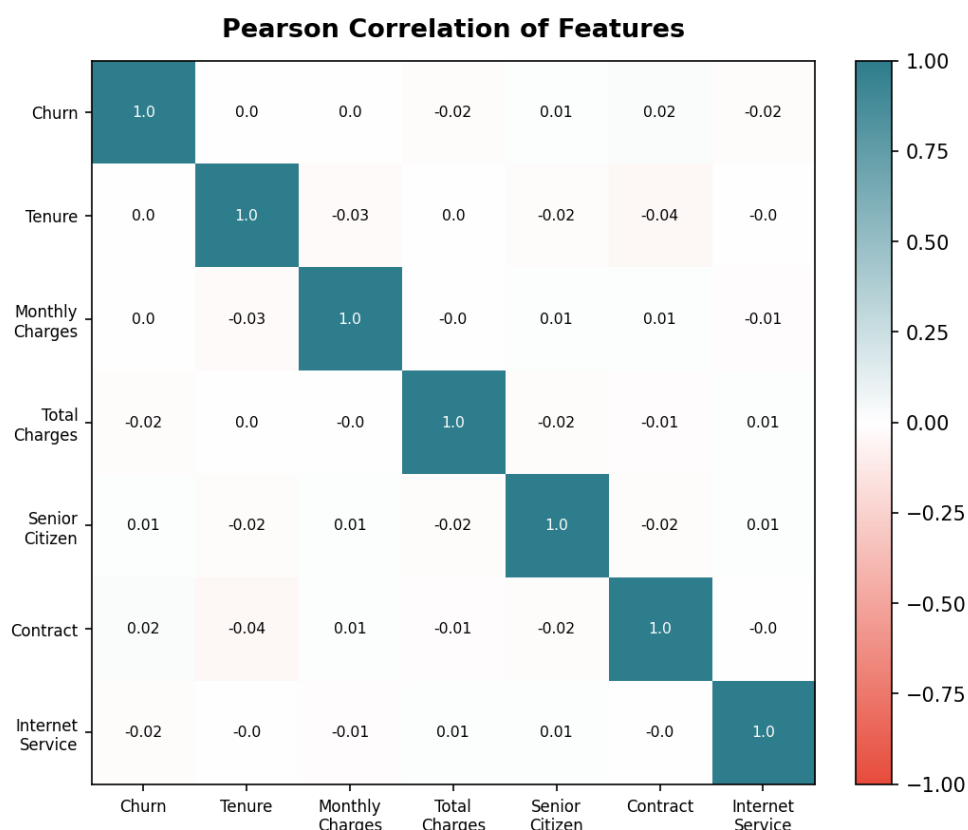


Figure 7 – Pearson correlation heatmap across key features.

6. Key Findings & Recommendations

- **Contract length is the strongest predictor.** Customers on month-to-month contracts churn at ~3× the rate of two-year contract holders. Incentivising annual/biennial commitments is the highest-ROI retention lever.
- **Early-tenure intervention is critical.** Churn risk peaks in months 0–12. A targeted onboarding and check-in programme in the first year could meaningfully reduce overall churn.
- **Fiber optic dissatisfaction.** Despite being the premium tier, fiber optic customers churn more. Net Promoter Score surveys and service-quality audits are recommended for this segment.
- **Electronic-check payers are high risk.** Migrating customers to auto-payment methods (bank transfer / credit card) could reduce churn and improve cash-flow predictability simultaneously.
- **Senior citizens need targeted care.** A dedicated support and loyalty programme for customers aged 65+ could address the higher churn rate observed in this cohort.

- **Modelling recommendation.** Given the ~27% churn rate, tree-based models (XGBoost, LightGBM) with SMOTE oversampling or scale_pos_weight adjustment are recommended as a baseline. Features to prioritise: tenure, contract type, monthly charges, and internet service.

This EDA provides the analytical foundation for building a robust churn prediction model. The next steps involve feature engineering (e.g., tenure buckets, interaction terms), model selection, and A/B-testing of retention interventions informed by model explanations (SHAP values).